

Jie Li

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Research Interests

My research focuses on high-performance computing (HPC) and AI infrastructure, with an emphasis on observability, resource management, and energy efficiency. Building on this systems foundation, my research agenda extends to AI agents for HPC operation and management, the security of agents acting within HPC environments, and the integration of HPC with quantum computing.

- **High-performance computing:** system monitoring, workload characterization, scheduling, disaggregated memory, and energy-aware resource management.
- **AI infrastructure:** LLM serving, KV cache and memory management, and performance and power characterization of AI workloads.
- **AI agents for HPC operations:** agent-assisted cluster operation and management, including ongoing development for the NSF REPACSS system.
- **AI agent security in HPC:** evaluating and constraining agent behavior in shared computing environments.
- **HPC and quantum computing:** quantum arithmetic and polynomial synthesis, with broader interests in hybrid quantum-classical workflows and HPC integration.

Education

Ph.D., Computer Science, Texas Tech University, Lubbock, TX May 2024

Dissertation: *Optimizing High-Performance Computing Systems: Insights from System Monitoring, Workload Management, and Scheduling Strategies*

Advisor: Prof. Yong Chen

M.S., Computer Science, Texas Tech University, Lubbock, TX August 2019

Thesis: *PIMS: A Lightweight Processing-in-Memory Accelerator for Stencil Computations*

Advisor: Prof. Yong Chen

B.A., Architecture, Huaqiao University, Xiamen, China May 2012

Academic Appointments and Research Experience

Texas Tech University, Lubbock, TX

Research Assistant Professor, Department of Computer Science July 2026 – Present

- Lead research in HPC and AI infrastructure, spanning observability, resource and data management, energy-aware computing, cyberinfrastructure security, and AI-driven system management.
- Initiated a research direction on AI agents for autonomous HPC operation and management, using REPACSS as a production testbed.
- Contribute to the operation, optimization, and research use of the NSF REPACSS cluster (\$12.25M), including system software integration, performance analysis, and user support.
- Supervise graduate and undergraduate researchers and develop proposals on secure, autonomous, and energy-efficient computing infrastructure.

Assistant Director, NSF Cloud and Autonomic Computing Center, TTU Site January 2026 – Present

- Coordinate faculty, student teams, industry members, and external collaborators in developing and reviewing research projects for this NSF Industry–University Cooperative Research Center.
- Help organize semiannual Industry Advisory Board meetings, including project planning, presentation preparation, and follow-up with industry participants.
- Mentor students preparing presentations, posters, and reports for industry review; support proposal development, center reporting, and cross-institutional coordination.

Postdoctoral Researcher June 2024 – June 2026

- Co-designed and built the NSF REPACSS cluster (\$12.25M) from the ground up, contributing to system architecture, deployment, operation, and performance optimization.

- Served as technical lead of the Data-Intensive Scalable Computing Laboratory, coordinating research activities, student projects, and collaborative system development.
- Co-PI on the NSF CICI proposal SHIELD (2025, not funded), which proposed layered defense mechanisms for open-science cyberinfrastructure.

Research Assistant

September 2019 – May 2024

- Across graduate study (2019–2024), published 15 peer-reviewed papers (7 first-author) in venues including CLUSTER, ISC, ICPP, MemSys, IEEE CLOUD, IEEE BigData, and IEEE Transactions on Computers.
- Developed and maintained MonSTer (later adopted by Dell's Omnia project), the Disaggregation-Aware Scheduler, and xBGAS simulation tools, supporting research in HPC monitoring, workload characterization, scheduling, and memory systems.
- Mentored students on HPC monitoring, data management, and workload analysis, including a master's thesis that led to a CLOUD'23 publication.

Lawrence Berkeley National Laboratory, Berkeley, CA

Graduate Student Intern

Summers 2021, 2022, and 2023

Mentors: Brandon Cook and Georgios Michelogiannakis, in John Shalf's group

- Designed and implemented pipelines integrating LDMS, DCGM, and Slurm telemetry from NERSC's Cori and Perlmutter supercomputers.
- Applied machine learning and deep learning to classify, characterize, and predict HPC job behavior from large-scale time-series telemetry.
- Led development of the open-source Disaggregation-Aware Scheduler; internship research resulted in first-author publications at ISC'23 and CLUSTER'24.

Grants and Proposals

REPACSS: Empowering Scientific Discovery through Renewable Energy Powered Advanced Computing Systems and Services

National Science Foundation, Category II | Submitted 2023 | **Funded** | Total award: \$12,250,000

Role: Contributor to proposal development and infrastructure implementation (PI: Yong Chen; Co-PI: Alan Sill). Contributed proposal sections on data center monitoring and remote control.

SHIELD: Strengthening High-Performance Infrastructure with Enhanced Layered Defense

NSF CICI: UCSS | Submitted 2025 | Not funded | Requested: \$600,000

Role: Co-PI (PI: Yong Chen). Contributed project vision and technical approach for HPC cybersecurity.

DA Vinci: An Integrated Data Collection, Automation, and Visualization Framework for HPC Systems

NSF Frameworks | Submitted 2020 | Not funded | Requested: \$1,000,000

Role: Lead contributor (PI: Yong Chen; Co-PI: Alan Sill). Led core framework design and research methodology.

Peer-Reviewed Publications

- [18] Yiwei Yang, Mujahid Al Rafi, Jesun Sahariar Firoz, Zhen Peng, Xi Wang, Kenza Bouasker, Sayan Ghosh, **Jie Li**, Daniel Wang, Dong Li, Hyeran Jeon, Kevin Barker, Nathan Tallent, and Luanzheng Guo. OCEAN: Open-Source CXL Emulation for Hyperscale Architecture and Networking. In *The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC'26)*, 2026. To appear.
- [17] Yongjian Zhao, **Jie Li**, Chenxu Niu, Alan Sill, and Yong Chen. Power-Centric Observability for HPC Systems: Design, Deployment, and Evaluation on REPACSS. In *Practice and Experience in Advanced Research Computing (PEARC'26)*, pages 1–8, 2026.
- [16] Chenxu Niu, Wei Zhang, **Jie Li**, Yongjian Zhao, Tongyang Wang, Xi Wang, and Yong Chen. TokenPowerBench: Benchmarking the Power Consumption of LLM Inference. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI'26)*, 40(38), pages 32582–32590, 2026. doi: 10.1609/aaai.v40i38.40535. (Acceptance rate: 4,167/23,680 = 17.6%)
- [15] **Jie Li**, George Michelogiannakis, Samuel Maloney, Brandon Cook, Estela Suarez, John Shalf, and Yong Chen. Job Scheduling in High Performance Computing Systems with Disaggregated Memory Resources. In *IEEE International Conference on Cluster Computing (CLUSTER'24)*, pages 297–309, 2024. doi: 10.1109/CLUSTER59578.2024.00033.

- [14] **Jie Li**, George Michelogiannakis, Brandon Cook, John Shalf, and Yong Chen. Scheduling and Allocation of Disaggregated Memory Resources in HPC Systems. In *IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW'24)*, pages 1202–1203, 2024. doi: 10.1109/IPDPSW63119.2024.00206. (Short paper)
- [13] **Jie Li**, John D. Leidel, Brian Page, and Yong Chen. Towards Cycle-accurate Simulation of xBGAS. In *International Conference on Computing, Networking and Communications (ICNC'24)*, pages 468–472, 2024. doi: 10.1109/ICNC59896.2024.10556078.
- [12] Tommy Dang, Ngan V. T. Nguyen, **Jie Li**, Alan Sill, and Yong Chen. Spiro: Order-Preserving Visualization in High Performance Computing Monitoring. In *International Symposium on Visual Computing (ISVC'23)*, pages 109–120. Springer, 2023. doi: 10.1007/978-3-031-47969-4_9.
- [11] **Jie Li**, Rui Wang, Ghazanfar Ali, Tommy Dang, Alan Sill, and Yong Chen. Workload Failure Prediction for Data Centers. In *IEEE 16th International Conference on Cloud Computing (CLOUD'23)*, pages 479–485, 2023. doi: 10.1109/CLOUD60044.2023.00064.
- [10] Cristiano E. Caon, **Jie Li**, and Yong Chen. Effective Management of Time Series Data. In *IEEE 16th International Conference on Cloud Computing (CLOUD'23)*, pages 408–414, 2023. doi: 10.1109/CLOUD60044.2023.00055. (Student mentee first author)
- [9] **Jie Li**, George Michelogiannakis, Brandon Cook, Dulanya Cooray, and Yong Chen. Analyzing Resource Utilization in an HPC System: A Case Study of NERSC's Perlmutter. In *International Conference on High Performance Computing (ISC'23)*, pages 297–316. Springer, 2023. doi: 10.1007/978-3-031-32041-5_16. (Acceptance rate: $21/78 = 26.9\%$)
- [8] Tommy Dang, Ngan V. T. Nguyen, **Jie Li**, Alan Sill, Jon Hass, and Yong Chen. JobViewer: Graph-based Visualization for Monitoring High-Performance Computing System. In *IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT'22)*, pages 110–119, 2022. doi: 10.1109/BDCAT56447.2022.00021.
- [7] Tommy Dang, Ngan Nguyen, Jon Hass, **Jie Li**, Yong Chen, and Alan Sill. The Gap between Visualization Research and Visualization Software in High-Performance Computing Center. In *The Gap between Visualization Research and Visualization Software (VisGap'21)*, 2021. doi: 10.2312/visgap.20211089.
- [6] Xi Wang, Antonino Tumeo, John D. Leidel, **Jie Li**, and Yong Chen. HAM: Hotspot-Aware Manager for Improving Communications With 3D-Stacked Memory. *IEEE Transactions on Computers*, 70(6):833–848, 2021. doi: 10.1109/TC.2021.3066982.
- [5] **Jie Li**, Ghazanfar Ali, Ngan Nguyen, Jon Hass, Alan Sill, Tommy Dang, and Yong Chen. MonSTer: An Out-of-the-Box Monitoring Tool for High Performance Computing Systems. In *IEEE International Conference on Cluster Computing (CLUSTER'20)*, pages 119–129, 2020. doi: 10.1109/CLUSTER49012.2020.00022. (Acceptance rate: $27/132 = 20.5\%$)
- [4] Ngan Nguyen, Jon Hass, Yong Chen, **Jie Li**, Alan Sill, and Tommy Dang. RadarViewer: Visualizing the Dynamics of Multivariate Data. In *Practice and Experience in Advanced Research Computing (PEARC'20)*, pages 555–556, 2020. doi: 10.1145/3311790.3404538. (Short paper)
- [3] Vung Pham, Ngan Nguyen, **Jie Li**, Jon Hass, Yong Chen, and Tommy Dang. MTSAD: Multivariate Time Series Abnormality Detection and Visualization. In *IEEE International Conference on Big Data (BigData'19)*, pages 3267–3276, 2019. doi: 10.1109/BigData47090.2019.9006559.
- [2] **Jie Li**, Xi Wang, Antonino Tumeo, Brody Williams, John D. Leidel, and Yong Chen. PIMS: A Lightweight Processing-in-Memory Accelerator for Stencil Computations. In *Proceedings of the International Symposium on Memory Systems (MemSys'19)*, pages 41–52, 2019. doi: 10.1145/3357526.3357550.
- [1] Xi Wang, Antonino Tumeo, John D. Leidel, **Jie Li**, and Yong Chen. MAC: Memory Access Coalescer for 3D-Stacked Memory. In *Proceedings of the 48th International Conference on Parallel Processing (ICPP'19)*, pages 1–10, 2019. doi: 10.1145/3337821.3337867. (Acceptance rate: $106/405 = 26.2\%$)

Preprints

- [3] **Jie Li**. Trusted Credentials, Untrusted Behavior: Benchmarking LLM-Agent Security in High-Performance Computing. *arXiv preprint arXiv:2607.18485*, July 2026. (Sole author)
- [2] **Jie Li**, Tongyang Wang, and Yong Chen. From Tensor Buffer to Distributed Memory Hierarchy: A Survey of KV Cache Management for LLM Serving. *arXiv preprint arXiv:2607.02574*, June 2026.

- [1] **Jie Li**, Brandon Cook, and Yong Chen. ARcode: HPC Application Recognition Through Image-encoded Monitoring Data. *arXiv preprint* arXiv:2301.08612, January 2023.

Manuscripts Under Submission or Revision

- [3] Ziqing Guo, **Jie Li**, Ziwen Pan, and Yong Chen. NNQA: Neural-Native Quantum Arithmetic for End-to-End Polynomial Synthesis. Submitted to AAAI-27.
- [2] Tongyang Wang, **Jie Li**, Chenxu Niu, and Yong Chen. ECHO: Online Control of CPU Frequency for Energy-Efficient High-Performance Computing.
- [1] Batuhan Sencer, Chenxu Niu, **Jie Li**, and Yong Chen. Slimming Models, Saving Watts: Understanding and Modeling the Impact of Knowledge Distillation on GPU Clusters.

Open-Source Research Software

- **MonSter** — Out-of-the-box HPC monitoring framework. Published at CLUSTER'20; adopted by Dell's Omnia project.
- **Disaggregation-Aware Scheduler** — Simulation framework for job scheduling in HPC systems with disaggregated memory. Used in CLUSTER'24 and IPDPSW'24 publications.
- **xBGAS REV-CPU** — Cycle-accurate xBGAS simulation extension based on REV-CPU. Developed with Tactical Computing Laboratories; published at ICNC'24.
- **xBGAS Runtime** — Lightweight runtime supporting global address space extensions for RISC-V-based xBGAS simulation. Developed with Tactical Computing Laboratories.

Teaching

Teaching interests: operating systems, parallel and high-performance computing, computer architecture, distributed and cloud computing, and systems for machine learning at the undergraduate and graduate levels; new graduate seminars on AI infrastructure and on AI agents for computing systems.

Invited Lecturer and Programming Project Designer

Fall 2025

Parallel Processing (graduate), Texas Tech University

- Delivered four invited lectures and designed programming projects for 27 students; topics included job scheduling, compilation and job submission, and OpenMP.

Research Mentoring

Graduate Students

- **Jaechang Kim**, M.S. student (ongoing). AI agents for REPACSS cluster operation and management, automating Warewulf, Ansible, Spack, and Slurm workflows.
- **Rupak Kadel**, Ph.D. student (Fall 2025 – Present). Improving HPC monitoring for anomaly and intrusion detection.
- **Cristiano Caon**, M.S. (Fall 2022). Data volume reduction and query optimization in time-series databases; independent study (CS7000) and master's thesis. Outcome: CLOUD'23 publication.
- **Aniruddh Sanjaysinh Chavda** (M.S.) and **Huyen Nguyen** (Ph.D.) (Spring 2021). Usage behavior analysis by clustering job accounting data; Advanced Operating Systems course project.
- **Ruonan Wu**, M.S. (Spring 2021). Job accounting data analysis for the Quanah cluster; Advanced Operating Systems course project.
- **Ashhrita Puradamane Balachandra**, M.S. (Spring 2020). Improving InfluxDB query performance; Advanced Operating Systems course project.

Undergraduate Students

- **Yusheng Han** and **Zachary Kay** (Spring 2022). Running HPC applications and analyzing performance on the RedRaider cluster; independent study (CS4000).
- **Casey Root** (Spring 2021). Monitoring queue status through the Slurm REST API; independent study (CS4000).

Presentations

- [6] “Integrated Data Collection and Visualization Framework for Data Centers based on Telemetry Model,” J. Li et al. NSF CAC Industry Advisory Board Conferences, 2021–2024 (4 presentations: Lubbock, TX, May 2021; Denton, TX, November 2022; Tucson, AZ, May 2023; Lubbock, TX, December 2024).
- [5] “Towards Cycle-Accurate Simulation of xBGAS,” J. Li and Y. Chen. Latch-Up 2024, Cambridge, MA, April 19, 2024.
- [4] “Workload Failure Prediction for Data Centers,” J. Li, R. Wang, G. Ali, T. Dang, A. Sill, and Y. Chen. IEEE 16th International Conference on Cloud Computing (CLOUD’23), Chicago, IL, July 6, 2023.
- [3] “Advanced Visualization and Data Analysis of HPC Cluster and User Application Behavior,” J. Li, N. Nguyen, G. Ali, T. Dang, Y. Chen, and A. Sill. ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC’21), November 2021.
- [2] “MonSTer: An Out-of-the-Box Monitoring Tool for High Performance Computing Systems,” J. Li, G. Ali, N. Nguyen, J. Hass, A. Sill, T. Dang, and Y. Chen. IEEE International Conference on Cluster Computing (CLUSTER’20), 2020.
- [1] “PIMS: A Lightweight Processing-in-Memory Accelerator for Stencil Computations,” J. Li, X. Wang, A. Tumeo, B. Williams, J. D. Leidel, and Y. Chen. International Symposium on Memory Systems (MemSys’19), October 2019.

Refereed Posters

- [3] J. Li, B. Cook, G. Michelogiannakis, and Y. Chen. A Holistic View of Memory Utilization on Perlmutter. SC’22, 2022.
- [2] J. Li, B. Cook, and Y. Chen. Detecting and Identifying Applications by Job Signatures. SC’21, 2021.
- [1] X. Wang, J. Li, A. Tumeo, J. D. Leidel, and Y. Chen. Memory Hotspot Optimizations for 3D-Stacked Memory. International Conference on Parallel Architectures and Compilation Techniques (PACT’19), 2019.

Awards and Honors

Best Poster Award, NSF Cloud and Autonomic Computing Center Industry Advisory Board Conference 2022
Summer Thesis/Dissertation Research Award (\$2,300), Texas Tech University Graduate School Summer 2019

Professional Service

- **Program committee:** NeurIPS 2026; AAAI 2026.
- **Conference reviewer:** IEEE ISCAS 2026; IEEE BigData 2022, 2023, 2025; IEEE/ACM CCGrid 2024.
- **Journal reviewer:** IEEE Computer Architecture Letters (2025); The Journal of Supercomputing (2023).
- **Conference sub-reviewer:** IEEE IPDPS 2023; IEEE ICDCS 2022; ACM/IEEE SC 2022; International Parallel Data Systems Workshop (PDSW) 2022; IEEE Smart Data Services 2020.
- **Student volunteer:** ACM/IEEE SC’21, St. Louis, MO; ACM/IEEE SC’19, Denver, CO.

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